Syllabus and Study guide for AI Mid Term exam

Sources

- Russell & Norvig's textbook (AIMA)
- Slides posted via Google classroom

Syllabus

Ch.	Subject	Topics/Skills required
1	Introduction	• Understand the four different approaches of AI • Know about the state-the-art applications of AI
2	Intelligent Agents	• Understand PEAS description • Understand the characteristics of environments (static/dynamic, stochastic/deterministic, etc.) • Understand different types of agents (model-based, utility based, etc.) • Be able to answer questions regarding the above from given scenario/agent description
3	Search (Blind + Heuristic)	• Understand the working principles and characteristics of all blind and heuristic search techniques (BFS, DFS, UCS, Greedy, A* etc.) • Solve mathematical problems (on graph/tree) related to these search techniques
5	Adversarial Search	• Understand game tree, minimax and alpha-beta pruning • Solve mathematical problems (on game tree) on minimax and alpha-beta pruning (https://www.javatpoint.com/ai-alpha-beta-pruning) • Understand techniques to make game tree search faster (cut-off search with heuristic evaluation function, move ordering, etc.)
6	CSP	• Understand CSP basics, backtracking, forward checking, heuristics to improve backtracking. • Use these to solve CSP problems